

ADVANCED METERING INFRASTRUCTURE (AMI) & SMART GRID DISTRIBUTION — REFERENCE ARCHITECTURE

STANDARDS: IEC 62443 (ZONES & CONDUITS) • IEC 62056 (DLMS/COSEM) • IEEE 2030.5 • PURDUE L0-L4 SEGMENTATION • DERMS INTEGRATION

[ZONE 1] FIELD EDGE & PREMISES (PURDUE L0/L1)

GRID CONNECTION POINTS • SUB-SECOND SAMPLING • 30-MIN INTERVAL NVRAM LOGGING
• BIDIRECTIONAL METROLOGY

Distributed Energy Resource (DER) / Solar Inverter

Protocol: Modbus-TCP / SunSpec • Port: 502
Profile: Base Allocation / Dynamic Operating Envelope (DOE)
Dynamic Curtailment & Volt/Var Support Active

Modular Type 4 Advanced Smart Meter

Unique Connection Identifier • IEC 62056-5-3 Compliant • Class 0.5
Metrology

Primary Metrology Core

Voltage (V), Current (A), Power Factor (PF), Frequency (Hz)
True RMS Sampling @ 4 kHz • 4-Quadrant (kW / kVAR)

Internal SPI Bus

Meter Controller & Cryptographic Core

DLMS/COSEM Object Dictionary • AES-128-GCM Session
Auth
90-Day Non-Volatile Memory (NVRAM) Storage Ring

Remote Service Disconnect Contactor (UC3)

Motorized Breaking Contactor • State: CLOSED
Safety Interlocks: Neutral Conductivity & Downstream Voltage
Check

Modular Swappable Transceiver Card

Dual Path: Cellular LTE Cat-M1 eSIM + Sub-GHz RF Mesh
Transit IP: Private APN Subnet / IPv6 6LoWPAN Mesh

ANSI/IEC Optical Maintenance Port (Protected Field
Calibration)

Facility Electrical Service & Distribution Board

Grid Connection: Low-Voltage 1-Phase / 3-Phase Main Switchboard

[ZONE 2] FIELD TELECOM & CARRIER WAN (PURDUE L2)

CARRIER VRF • ENCRYPTED IPSEC BACKHAUL • MULTI-BEARER REDUNDANCY

Cellular LTE Cat-M1 Private APN Gateway

Private APN Gateway • Dedicated Carrier VRF
Zero Public Internet Exposure • QoS Priority Class 1

Private VRF APN Transit

Wi-SUN Mesh Field Area Network (FAN)

IEEE 802.15.4g Sub-GHz RF Mesh Collector Gateway
Pole-Top Border Router -> High-Speed Fiber / WAN Backhaul

Substation WAN Backhaul

Carrier Interconnect IPsec Concentrator

Tunnel: IKEv2 / IPsec ESP (AES-256-GCM / SHA-384)
BGP Peering • Dual Redundant Cryptographic Endpoints

Real-Time High-Priority Event Bus (Distributed)

Last-Gasp Outage Alarms (<2 sec SLA) • Tamper Alarms
High-Throughput Message Queue Partitioned by Endpoint ID

Bulk Interval Telemetry Ingestion Buffer

DLMS/COSEM Push Messages & Scheduled Sweeps
Scalable Clustered Ingestion Buffer Architecture

[ZONE 3] AMI HEAD-END OPERATIONS & IDMZ (PURDUE L3/L3.5)

Dynamic Curtailment Broadcast (DOE Curve) • GEN FW • VEE ENGINE • MDMS VAULT

IDMZ Next-Gen Perimeter Firewalls (HA Pair)

Stateful Inspection • DLMS Protocol Deep Packet Inspection
Port 4059 Ingress Only • Zero Trust Network Access (ZTNA)

Port 4059 DLMS Ingress (VLAN 100)

AMI Head-End System (HES) FEP Cluster

DLMS/COSEM HES FEPs • AMI Head-End • Device Lifecycle & Discovery
HSM Key Management Vault • AES-128 GCM Session Broker

High-Speed Telemetry Stream

VEE Engine (Validation, Editing & Estimation)

Metrology Compliance Procedures • Gap Detection &
Interpolation
Suspect Data Flagging • High-Precision Clock Drift
Synchronization

Validated Interval Records (VLAN 200)

MDMS Historical Master Database Cluster

High-Performance Timescale/Relational Database Cluster
Master Meter Records • Multi-Year Interval Profiles • Market
Aggregator

Remote Service Order Command Dispatcher

Cryptographically Signed Command Queue • Audit Trail Engine
Automated Interlock Safety Proof Verification before Switching

AMI Core L3 10GbE Switch Fabric (VLAN Segmentation)

VLAN 100 (Ingress) • VLAN 200 (MDMS DB) • VLAN 300 (DERMS) • VLAN 400
(B2B)

[ZONE 4] ENTERPRISE, DERMS & MARKET INTERFACES (PURDUE L4)

DISTRIBUTION GRID OPERATIONS • DYNAMIC DER MANAGEMENT • WHOLESALE MARKET • CONSUMER
PORTAL

DERMS & Dynamic Grid Export Controller

Distributed Energy Resource Management System
Dynamic Operating Envelopes (DOE) • Transformer Thermal Limits
Low-Voltage Reverse Power Flow Mitigation Algorithms

ADMS / Outage Management System (OMS)

Advanced Distribution Management System
Low-Voltage (LV) Topology Visibility & Phase Load Balancing
Automated Fault Scope Pinpointing via Last-Gasp Telemetry

Energy Market Operator (EMO) B2B Gateway

Wholesale Market Settlement Interface (Standard XML Schemas)
Retailer B2B Service Orders: Remote Disconnect / Reconnect
Standard Transaction Data Feeds & Market Participant Gateway

Consumer Web Portal & Mobile App Gateway

Consumer Interval Consumption & Solar Export Visualizer
Dynamic Time-of-Use Signals • Outage & Restoration Tracker

OT Security Operations Center (SOC) & SIEM

Enterprise OT SIEM Platform • Tamper Anomaly Alarms
NetFlow, Syslog & TLS Handshake Telemetry Ingestion

Syslog & NetFlow Mirror